

PRD-013 — Question Content System

Status

Approved

1. Purpose

The Question Content System is responsible for creating, storing, organizing, validating, previewing, and publishing every SAT question used throughout PrepHub.

It serves as the single source of truth for all assessment content, including:

- Diagnostic questions
- Adaptive practice questions
- Question families (replicas)
- Video explanations
- Written explanations
- Images
- Answer validation
- Question metadata

Every question presented to a student originates from this system.

2. Product Philosophy

The Question Content System should prioritize:

- Content quality over upload speed
- Consistency across all questions
- A simple workflow for the PrepHub owner
- Prevention of incomplete or broken questions
- Long-term scalability as the question bank grows

Students should never experience the complexity of the content management system. Their experience should always feel polished and seamless.

3. Goals

The system shall:

- Store all SAT questions
- Support both SAT question types
- Organize questions into categories
- Support manually created question families (replicas)
- Require video explanations
- Allow optional written explanations
- Support mathematical notation throughout
- Support diagrams and images
- Require owner preview before publishing
- Prevent incomplete questions from being published

4. Supported Question Types

PrepHub supports two question types.

Multiple Choice

Each question contains:

- Question text
- Four answer choices
- Exactly one correct answer

Every multiple-choice question must contain exactly four answer choices before it can be published.

Open-Ended Numeric Response

Students type a numeric answer.

The owner selects this question type during question creation.

Open-ended questions do not contain answer choices.

5. Data Model

The Question Content System is organized around ****Question Families****.

A Question Family represents one underlying SAT concept and contains:

- One shared video explanation
- Up to three manually created question versions (replicas)

Each Question Version is an individual question presented to students.

This architecture allows multiple versions of the same concept while sharing one explanation video.

6. Question Family

Every Question Family contains exactly ****three manually created versions****.

Each version represents the same underlying concept but may differ in:

- Numbers
- Variables
- Wording
- Answer choices

Students never know that multiple versions belong to the same family.

Each family stores:

- Shared video explanation
- Creation date
- Published status
- Analytics aggregated across the family

7. Question Version

Each Question Version stores its own independent information.

Required fields:

- Question type

- Category
- Difficulty
- Question text
- Suggested time
- Calculator setting
- Correct answer(s)

Optional fields:

- Written explanation
- One question image

Question Versions inherit the shared video explanation from their Question Family.

8. Categories

Every Question Version belongs to exactly one category.

Available categories:

- Reading Comprehension
- Grammar
- Vocabulary
- Algebra
- Geometry & Trigonometry
- Advanced Math
- Problem Solving & Data Analysis

Questions cannot belong to multiple categories.

PrepHub V1 does not use subtopic tagging.

9. Difficulty

Every Question Version has one manually assigned difficulty.

Available values:

- Easy
- Medium

- Hard

Difficulty is assigned during upload.

PrepHub does not automatically change difficulty.

10. Video Explanations

Every Question Family has one required video explanation.

The owner uploads the video directly into PrepHub.

The video explains the solving process generally so that it applies to all three versions within the family.

A Question Family cannot be published without a video.

11. Written Explanations

Written explanations are optional.

If provided, they appear after the student submits the question alongside the video explanation.

Questions may be published without a written explanation.

12. Mathematical Formatting

PrepHub supports full LaTeX rendering throughout the platform.

LaTeX is supported in:

- Question text
- Multiple-choice answer choices
- Written explanations

The renderer must support all SAT mathematical notation including:

- Fractions

- Exponents
- Square roots
- Radicals
- Greek letters
- Subscripts
- Superscripts
- Absolute values
- Inequalities
- Piecewise notation
- Summations
- Integrals
- Matrices
- Any other mathematical notation commonly required on the SAT

13. Images

Each Question Version may contain:

- One optional question image

If the question is multiple choice:

Each answer choice may also contain:

- One optional image

Open-ended questions do not have answer-choice images.

14. Multiple Choice Questions

Every multiple-choice question contains exactly four answer choices.

Each answer choice contains:

- Text
- Optional image

Answer choice text fully supports LaTeX.

Exactly one answer choice is marked as correct.

15. Open-Ended Answer Validation

For open-ended questions, the owner may specify multiple accepted answers.

Example:

- 0.5
- .5
- 1/2

Student answers are normalized for basic formatting (such as trimming whitespace) before comparison.

The submitted answer must exactly match one of the accepted answers.

PrepHub does not support numerical tolerance.

16. Calculator Setting

Every Question Version specifies whether a calculator is allowed.

Options:

- Calculator Allowed
- Calculator Not Allowed

This determines whether the integrated calculator is available to students.

17. Suggested Time

Every Question Version includes a recommended completion time.

This value is entered manually by the owner.

Students see this recommended time while answering the question.

If the recommended time expires, students are informed but may continue working without penalty.

18. Question Preview

Before publishing, the owner must preview the question.

The preview displays the question exactly as a student will see it, including:

- Question text
- Images
- LaTeX rendering
- Answer choices
- Calculator availability
- Open-ended answer input

Publishing is not permitted until the preview has been viewed.

19. Publishing Workflow

Every Question Family follows the workflow:

Draft

↓

Edit

↓

Mandatory Student Preview

↓

Publish

Published questions may be edited at any time.

Edits become live immediately.

20. Publishing Checklist

Publishing is blocked until every required field has been completed.

Required:

- Question text
- Category
- Difficulty
- Question type
- Calculator setting
- Suggested time
- Video uploaded
- Correct answer(s)
- Four answer choices (if multiple choice)

Optional:

- Written explanation
- Question image
- Answer choice images

21. Analytics

PrepHub records analytics for every published Question Version.

Available analytics include:

- Times served
- Correct response percentage
- Incorrect response percentage
- Average completion time
- Answer choice distribution
- Video explanation views

Analytics are visible only within the Owner Dashboard.

22. Editing

The owner may edit any Question Family or Question Version at any time.

Changes to published questions become live immediately.

PrepHub V1 does not maintain version history.

23. Unpublishing

The owner may unpublish any Question Family immediately.

Unpublished questions are no longer eligible for future practice sets.

Historical student results remain unchanged.

PrepHub does not recalculate previous predictions or results after a question is unpublished.

24. Deletion

The owner may permanently delete any Question Family.

Deletion permanently removes all associated Question Versions from PrepHub.

This action cannot be undone.

25. Permissions

Only the PrepHub Owner Account may:

- Create Question Families
- Create Question Versions
- Edit questions
- Upload videos
- Upload images
- Preview questions
- Publish questions
- Unpublish questions
- Delete questions

No additional content editor roles exist in V1.

26. Acceptance Criteria

The Question Content System is complete when:

- Multiple Choice and Open-Ended Numeric Response questions are fully supported.
- Every question belongs to exactly one category.
- Every question has one manually assigned difficulty.
- Every Question Family contains exactly three manually created versions.
- Every Question Family has one required shared video explanation.
- Written explanations remain optional.
- Full LaTeX rendering works throughout question text, answer choices, and written explanations.
- Questions support one optional image.
- Multiple-choice answer choices support one optional image each.
- Open-ended questions support multiple exact accepted answers.
- Numerical tolerance is not supported.
- Calculator permission is configurable per question.
- Suggested time is configurable per question.
- Mandatory preview is required before publishing.
- Publishing is blocked until all required fields are complete.
- Question analytics are recorded.
- The owner can edit, unpublish, and permanently delete content.
- Only the Owner Account has access to the Question Content System.